

ClimaXtreme

Extreme Climate Analysis and Countermeasures

In extreme - climate research, moments come when you see the need for action so clearly that you want to turn ideas into reality.

Group Member: YIN Hanyu, DU Yu, CHEN Zitao

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queries for various regions

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Measures for different
extreme weather events

C O N T E N T S

PART 01 | Background & Homepage



Background: Reason for choosing this topic

As global climate change gets worse, the frequency and intensity of extreme weather events have increased a lot. Most of the weather services focus on accurate data reporting and lack the shift from data to practical action guides.

Our website includes both daily prevention and emergency response situations, meets the needs of different kinds of users, and hopes to provide ordinary people with quick and easy emergency plans for extreme weather and also real-time information for professionals





Usage scenarios

Daily prevention

Users could get future extreme weather forecasts, learn about potentially affected areas and times in advance, and study science knowledge and protection measures for different types of extreme weather.

Emergency response

After the government issues extreme weather warnings, users can quickly check targeted response measures and view real-time disaster situation updates. They can also use the AI assistant to find response measures quickly and get targeted risk avoidance tips.

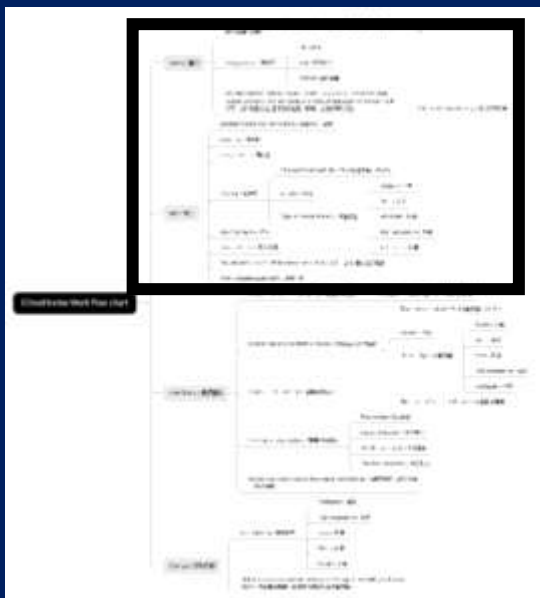
Users can search for relevant disaster news, reduce information gaps, and learn about the latest updates of affected areas.

Audience analysis

- General Public: The audience of our website needs intuitive and easy-to-use pages, and the AI assistant can help them solve problems.
- Researchers & Analysts: Our website uses various aspects of chart data and disaster distribution to help users understand disaster response measures.
- Policy Makers: We may need an overall disaster statistical panel to assess risk trends and develop disaster prevention strategies.

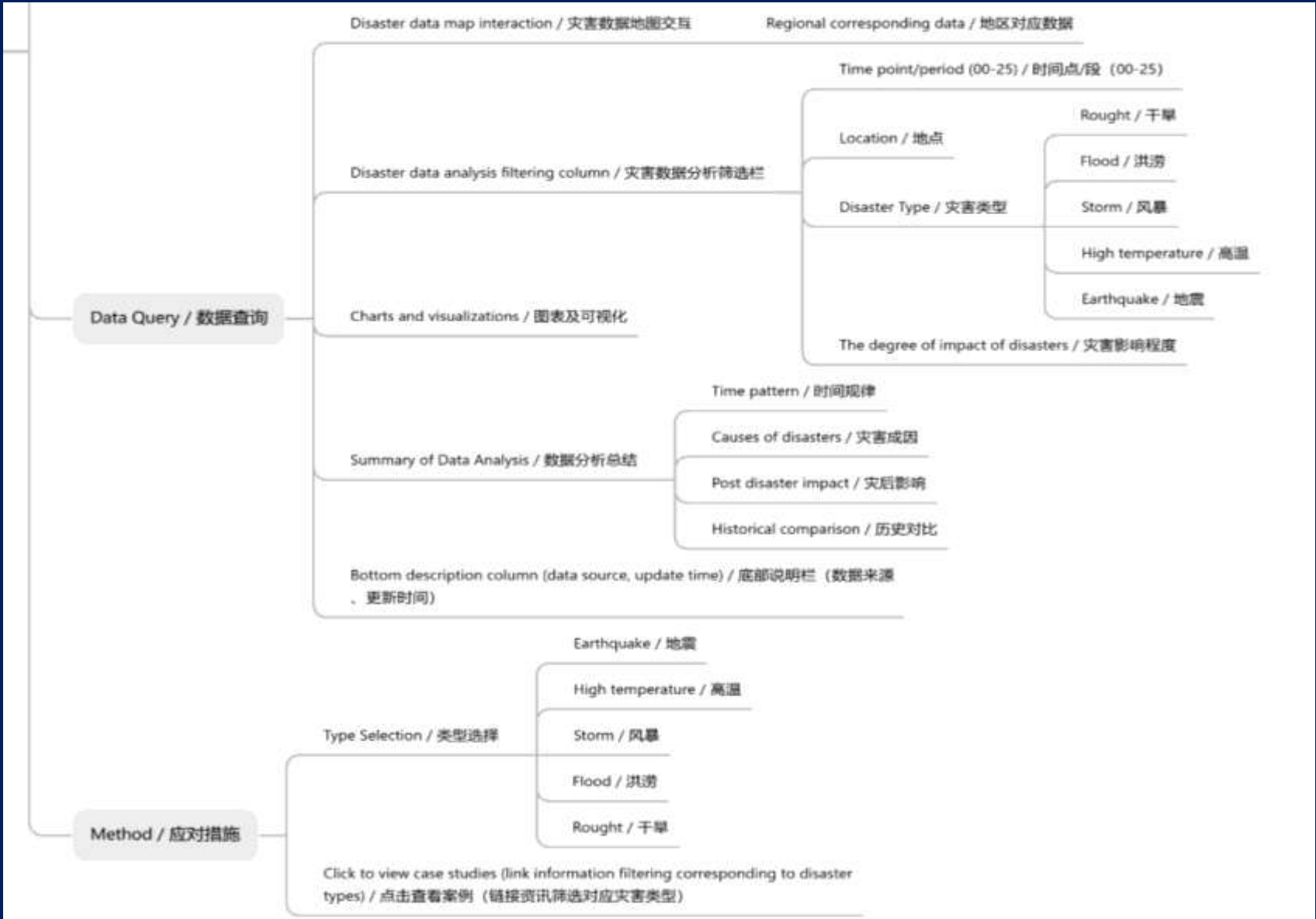
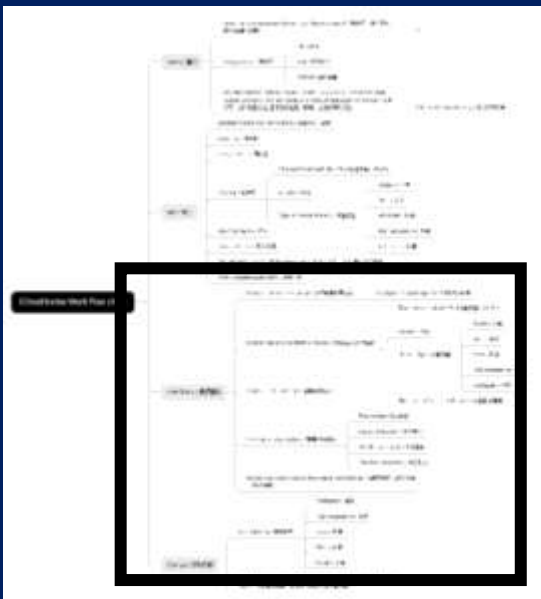


Website Workflow Chart





Website Workflow Chart





Homepage Design

ClimaXtreme



ClimaXtreme

Moving the mouse over the title will display a faint glow for a week.



The background is a city with black outlines and bright yellow lights, which look like stars shining under the night sky from afar, providing guidance for people in extreme weather conditions.



weather conditions



Main page display

Obtain weather information. Important data such as temperature, humidity, and precipitation probability are clearly displayed.

By positioning

ClimaXtreme With precise positioning technology, it can display the weather conditions in your location in real time.

By search

simply enter the place name in the search box to get the weather information for different area.





AI assistant



AI summarizes climate characteristics

You can access the AI Q&A interface by clicking the search box in the upper right corner of the main page. It summarizes global climate characteristics, helping users quickly and easily understand extreme weather



C O N T E N T S

PART 02 | Data visualization



Data overview



Page positioning

explore disaster characteristics in depth under different dimensions through screening criteria.



Data filter

[provides us with a flexible analytical perspective, enabling us to quickly obtain detailed data



Overview

Total number of events and total economic losses are important data that can help users understand disaster patterns



Page navigation

When analyzing further from the perspectives of "spatial distribution" or "trend changes," simply click to switch to present the data in richer visualization formats.



data visualization



Disaster Map

Focusing on the spatial distribution of disasters: A map of China is selected for different disaster types, with the severity of disasters in each province marked by a blue-white gradient, and color-coded scale bars provided for easier interpretation.



Data Charts

Presents changes in the number of disaster events, the distribution of types, the frequency of disasters in multiple provinces, and a dual-axis line chart that simultaneously displays the trends in the number of deaths and economic losses.

C O N T E N T S

PART 03 | Data Query Page (Database-Based)



Backend implementation (based on database)

```
spring.datasource.url=jdbc:sqlite:E:/AIDM7450/AIDM7450_GroupWork/DataBase/wether_information.db.db
spring.datasource.driver-class-name=org.sqlite.JDBC
spring.datasource.username=
spring.datasource.password=
#debug=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.SQLiteDialect
```

Database configuration file for the Spring Boot project

```
public List<DisasterData> getByStartYear(Integer startYear) {
    if (startYear == null || startYear < 2000 || startYear > 2025) {
        throw new IllegalArgumentException("无效的开始年份: 必须是2000-2025之间的整数");
    }
    List<DisasterData> disasters = disasterRepository.findByStartYear(startYear);
    if (disasters != null) {
        disasters = disasters.stream()
            .filter(d -> d.getStartYear() != null)
            .collect(Collectors.toList());
    }
    return disasters != null ? disasters : List.of();
}
```

Business logic method for querying disaster data by start year in the Spring Boot backend

```
public List<DisasterData> findByStartYear(Integer startYear) {
    String sql = "SELECT \"DisNo.\" AS disNo, \"Historic\", \" +
        \"\"Classification Key\", \"Classifier Group\", \" +
        \"\"Disaster Type\", \"Country\", \"Location\", \" +
        \"\"Start Year\", \"Total Damage ('000 US$)\" \" +
        \"AS totalDamage, \"Total Affected\", \"Magnitude\" \" +
        \"FROM disaster_data WHERE \"Start Year\" =?\";
    return jdbcTemplate.query(
        sql,
        new BeanPropertyRowMapper<>(DisasterData.class),
        startYear
    );
}
```

Method in the Repository (Dao) layer of the Spring Boot backend data access layer

```
@GetMapping("/by-year")
public ResponseEntity<List<DisasterData>> getByStartYear(
    @RequestParam Integer year // 接收前端传递的年份参数
) {
    List<DisasterData> data = disasterService.getByStartYear(year);
    return new ResponseEntity<>(data, HttpStatus.OK);
}
```

Controller interface method in the Spring Boot backend



WebPage Display - 1

灾难数据查询

查询所有灾难数据 获取全部数据

按灾害编号查询 (如: 2012-0173-CHN)
输入DisNo.
查询

按开始年份查询 (如: 2000-2025)
输入年份
查询

按地址查询 (如: guangdong)
输入地址关键词
查询

按灾害类型查询 (如: Flood)
输入灾害类型
查询

灾害编号	灾害类型	国家	发生地点	开始年份	总损失(千美元)	受影响人数	震级	操作
------	------	----	------	------	----------	-------	----	----

The header section of a disaster information management table:

- Disaster ID
- Disaster Type
- Country
- Location
- Start Year
- Total Loss
- Affected Population
- Magnitude
- Actions

Initial State Page of the Disaster Data Query Page



WebPage Display - 2

查询所有灾难数据

获取全部数据

按灾害编号查询 (如: 2012-0173-CHN)

输入DisNo.

查询

按开始年份查询 (如: 2000-2025)

输入年份

查询

按地址查询 (如: guangdong)

输入地址关键词

查询

按灾害类型查询 (如: Flood)

输入灾害类型

查询

- **Query All Disaster Data:** Click the "Get All Data" button to view the full dataset.
- **Query by Disaster ID:** Enter the ID and click "Query" to locate a single disaster record (this is an exact query).
- **Query by Start Year:** Enter a year between 2000 and 2025 to filter disasters from that year; a reminder will pop up if the entered year is outside this range.
- **Query by Location:** Enter a regional keyword to match records where the occurrence location contains the keyword (this is a fuzzy query).
- **Query by Disaster Type:** Enter a disaster type to filter disasters of the corresponding category (e.g., earthquakes, storms, floods, etc.).





WebPage Display - 3

灾难数据查询

查询所有灾难数据

获取全部数据

按灾害编号查询 (如: 2012-0173-CHN)

输入DisNo.

查询

按开始年份查询 (如: 2000-2025)

输入年份

查询

按地址查询 (如: guangdong)

输入地址关键词

查询

按灾害类型查询 (如: Flood)

输入灾害类型

查询

灾害编号	灾害类型	国家	发生地点	开始年份	总损失(千美元)	受影响人数	震级	操作
2000-0033-CHN	Earthquake	China	Yongren Xian, Yaolan Xian, Nanhua Xian, Dayao Xian areas (Chuxiong Yi district, Yunnan Sheng province), Xiangyun Xian area (Dali Bai district, Yunnan Sheng province)	2000	73500	1855007	5.9	操作
2000-0036-CHN	Earthquake	China	Mile Xian area (Honghe Hani and Yi district, Yunnan Sheng province), Qubei Xian area (Wenshan Zhuang and Miao district, Yunnan Sheng province)	2000	483	10302	4.9	操作
2000-0041-CHN	Earthquake	China	Xuyan Manzu Zizhilian area (Amshan ...	2000	无	62030	5.1	操作
2000-0156-CHN	Storm	China	Qi Xian area (Jiuzhong district, Shanxi ...	2000	无	53	79	操作
2000-0163-CHN	Storm	China	Ik Kazakh district (Xinjiang Uygur Zibh...	2000	3100	18004	无	操作
2000-0200-CHN	Flood	China	Shenzhen, Zhuhai, Zhongshan district...	2000	12000	40000	21000	操作
2000-0270-CHN	Storm	China	Guangdong Sheng, Guangxi Zhuangz...	2000	5000	无	无	操作
2000-0298-CHN	Earthquake	China	Lanzhou Shi area (Lanzhou district, Ga...	2000	无	100	5.6	操作
2000-0302-CHN	Flood	China	Awu, Jingmu, Hadapu villages (Dangc...	2000	22000	154183	114000	操作
2000-0326-CHN	Mass movement (we)	China	Sichuan Sheng, Guizhou Sheng provin...	2000	无	2000	无	操作
2000-0330-CHN	Flood	China	Sichuan Sheng, Yunnan Sheng provinc...	2000	无	15000	无	操作
2000-0350-CHN	Storm	China	Quanzhou district (Fujian Sheng provi...	2000	无	10	无	操作
2000-0372-CHN	Flood	China	Quanzhou, Zhangzhou, Xiamen distric...	2000	19000	31010	76000	操作
2000-0376-CHN	Flood	China	Duyun Shi area (Qiannan Buyei and M...	2000	60000	300000	无	操作
2000-0425-CHN	Storm	China	Baoying Xian, Gaoyou Shi, Yangzhou S...	2000	400	20500	无	操作
2000-0426-CHN	Flood	China	Bayu Xian area (Garze Tibetan district...	2000	无	无	无	操作
2000-0472-CHN	Flood	China	Hu Xian (Q'an district, Shaanxi Sheng ...	2000	172000	18214	无	操作

The page displayed after clicking the "Get All Data" button



WebPage Display - 4

灾难数据查询

查询所有灾难数据 获取全部数据

按灾难编号查询 (如: 2012-0173-CHN)

2002-0481-CHN

查询

按开始年份查询 (如: 2000-2025)

输入年份

查询

按地址查询 (如: guangdong)

输入地址关键词

查询

按灾害类型查询 (如: Flood)

输入灾害类型

查询

灾难编号	灾害类型	国家	发生地点
2002-0481-CHN	Flood	China	Shaoguan, Meizhou, Qingyuan

- Querying disaster information by disaster number is an exact search.
- The entered disaster number is case-sensitive, empty values are not accepted, and an error will be reported if an invalid disaster number is entered.
- The input box will be cleared after clicking the query button.

The page displayed after clicking the "Query by Disaster ID" button





WebPage Display - 5

灾难数据查询

查询所有灾难数据 [获取全部数据](#)

按灾害编号查询 (如: 2012-0173-CHN)

[查询](#)

按开始年份查询 (如: 2000-2025)

[查询](#)

按地址查询 (如: guangdong)

[查询](#)

按灾害类型查询 (如: Flood)

[查询](#)

灾害编号	灾害类型	国家	发生地点
2010-0036-CHN	Storm	China	Xinjiang Uygur Zizhiqu provin
2010-0064-CHN	Earthquake	China	Moxizhen village (Sulining dis
2010-0087-CHN	Earthquake	China	Kunming district (Yunnan She
2010-0169-CHN	Earthquake	China	Yushu Xian area (Name Unkno
2010-0179-CHN	Storm	China	Chongqing Shi province
2010-0216-CHN	Mass movement (wet)	China	Guangxi Zhuangzu Zizhiqu pr
2010-0239-CHN	Flood	China	Ankang, Hanzhong, Shangluc
2010-0262-CHN	Mass movement (wet)	China	Dazhai village (Guanling Bouy
2010-0306-CHN	Storm	China	Hainan Sheng province
2010-0344-CHN	Flood	China	Jilin Sheng province
2010-0354-CHN	Mass movement (wet)	China	Hanyuan Xian area (Yi'an dist
2010-0377-CHN	Mass movement (wet)	China	Zhouqu Xian area (Gannan Ti

- Querying disaster information by the year of disaster occurrence is an exact search.
- The entered disaster year must be a positive integer, empty values are not accepted, and it must fall within the range of 2000 to 2025; otherwise, an error will be reported.
- The input box will be cleared after clicking the query button.

The page displayed after clicking the "Query by Year" button





WebPage Display - 6

灾难数据查询

查询所有灾难数据 获取全部数据

按灾难编号查询 (如: 2012-0173-CHN)
输入DisNo

查询

按开始年份查询 (如: 2000-2025)
输入年份

查询

按地址查询 (如: guangdong)
sichuan

查询

按灾害类型查询 (如: Flood)
输入灾害类型

查询

灾害编号	灾害类型	国家	发生地点
2000-0302-CHN	Flood	China	Awu, Jingmu, Hadapu village
2000-0326-CHN	Mass movement (wet)	China	Sichuan Sheng, Guizhou Sheng
2000-0330-CHN	Flood	China	Sichuan Sheng, Yunnan Sheng
2000-0426-CHN	Flood	China	Baiyu Xian area (Garze Tibetan
2001-0064-CHN	Earthquake	China	Yajiang Xian, Kangding Xian a
2001-0214-CHN	Earthquake	China	Ninglang Yi Zizhixian area (L
2001-0526-CHN	Storm	China	Panzhihua district (Sichuan S
2001-0534-CHN	Flood	China	Jinyang area (Liangshan Yi di
2001-0540-CHN	Mass movement (wet)	China	Mingshan Xian area (Ya'an di
2001-9216-CHN	Drought	China	Sichuan Sheng, Yunnan Sheng
2002-0350-CHN	Flood	China	Foping Xian, Yang Xian, Ning
2002-0599-CHN	Storm	China	Jiangxi Sheng, Sichuan Sheng

- Querying disaster information by the geographic location of disaster occurrence is a fuzzy search.
- The entered geographic location is not case-sensitive, but empty values are not accepted.
- The input box will be cleared after clicking the query button.

The page displayed after clicking the "Query by Location" button





WebPage Display - 7

灾难数据查询

查询所有灾难数据 获取全部数据

按灾难编号查询 (如: 2012-0173-CHN)

输入DisNo

查询

按开始年份查询 (如: 2000-2025)

输入年份

查询

按地址查询 (如: guangdong)

输入地址关键词

查询

按灾害类型查询 (如: Flood)

Storm

查询

灾害编号	灾害类型	国家	发生地点
2000-0156-CHN	Storm	China	Qi Xian area (Jinzhong district)
2000-0163-CHN	Storm	China	Bi Kazakh district (Xinjiang Uy)
2000-0270-CHN	Storm	China	Guangdong Sheng, Guangxi
2000-0350-CHN	Storm	China	Quanzhou district (Fujian She
2000-0425-CHN	Storm	China	Baoying Xian, Gaoyou Shi, Yai
2000-0532-CHN	Storm	China	Nanzheng Xian, Liuba Xian ar
2000-0533-CHN	Storm	China	Leping Shi area (Wenzhou dis
2000-0556-CHN	Storm	China	Xiangshui Xian area (Yanchen
2000-0557-CHN	Storm	China	Hunan Sheng, Guangdong Sh
2000-0605-CHN	Storm	China	Hainan Sheng province
2001-0008-CHN	Storm	China	Xin Gol, Xing'an, Hulunbuir
2001-0292-CHN	Storm	China	Fujian Sheng, Zhejiang Sheng

- Querying disaster information by disaster type is an exact search.
- The entered disaster type is case-sensitive and empty values are not accepted.
- The input box will be cleared after clicking the query button.

The page displayed after clicking the "Query by Disaster Type" button



C O N T E N T S

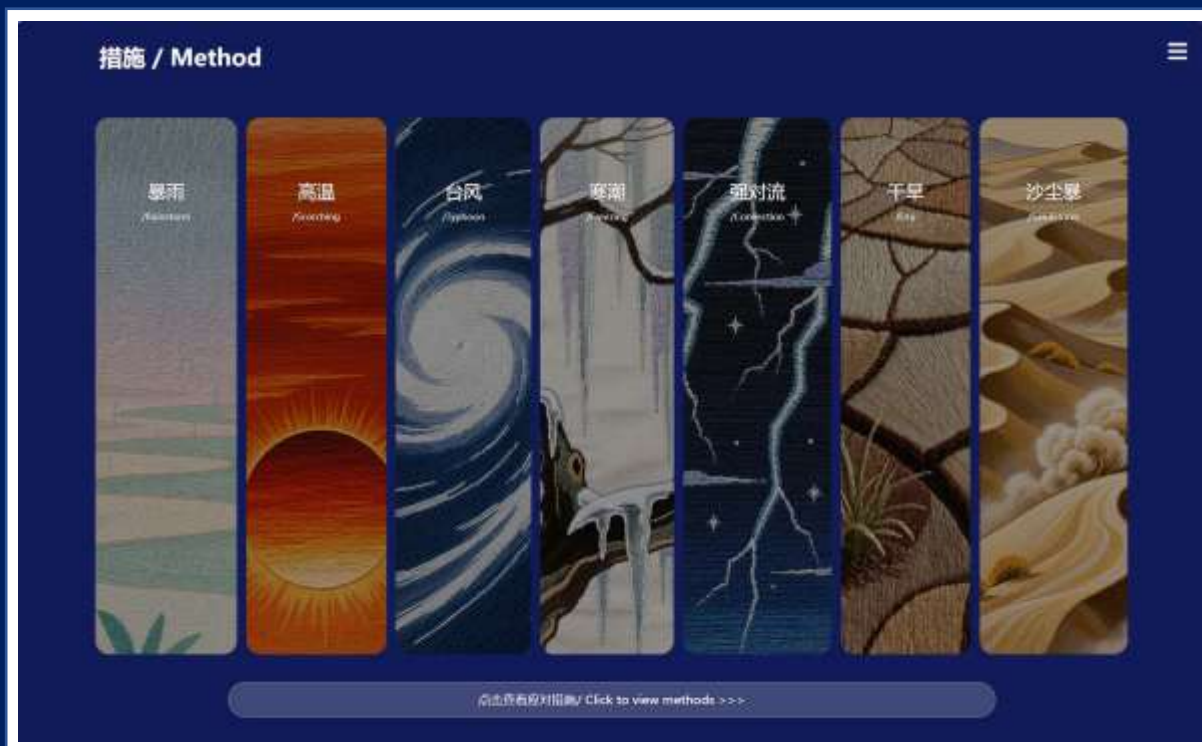
PART 04 | Methods



Method tags



Click “Method” and turn to this page.



- The main purpose of this page is to obtain relevant defense information in the fastest way possible during extreme weather events.
- The page displays 7 of the most common extreme weather for people to choose.
- At the bottom of the page, people can search for relevant news when they have free time.
- Tags will randomly be arranged according to the size of the page.



Method details



Click BG picture and back to tags.
Choose a tag and turn to this page.

措施 / Method



- 关闭门窗，用湿毛巾堵门缝/窗缝，减少沙尘进入；室内用空气净化器等。
- 外出必须戴好口罩（防沙尘吸入）和防风镜，不要露脸部位。
- 不骑自行车/电动车，驾车减速慢行，开雾灯+示廓灯，保持安全车距。
- 外出返回后，立即清洗面部、口罩，更换衣物，避免沙尘接触引发过敏。
- 群聚高危人群（哮喘/慢病患者）减少外出，避免病情加重。

点击查看应对措施 / Click to view methods >>>

This page aims to provide users with the fastest emergency plan for extreme weather.



Related news



Click news query and input a kind of weather.



- This page aims to provide people with corresponding types of news in their spare time.
- The search connection used still has the same functionality as the previous AI connection.
- AI will search for news related to the corresponding topic in the network and provide webpage links for direct redirection.

Thanks !

AIDM 7450 - Dynamic Web & Mobile Programming for Digital Media - ClimaXtreme

Group Member: YIN Hanyu, DU Yu, CHEN Zitao